Extended results and processing details

Sample Efficiency of AlphaEarth Embeddings for Land Cover Classification: Spatial Holdout and Cross-Country Evaluation

S1. Reference data selection and feature extraction

Table S1 gives the number of LUCAS 2018 Copernicus polygons after each selection step, counted in Google Earth Engine. All features were extracted with the Earth Engine reduceRegions operation and a mean reducer at a nominal scale of 10 m. Because the mosaicked collections carry no fixed projection, the reduction was evaluated on a 10 m grid in geographic coordinates (EPSG:4326); every grid cell took the value of the native 10 m pixel at its location (nearest neighbour), and cells were weighted by the fraction of their area inside the polygon. For polygon 22543 (1,566 m^2), for example, 20 grid cells had their centre inside the polygon and the summed fractional weights were 20.47. The mean therefore approximates an area weighted mean of the native pixels. The polygons contained 3 to 77 grid cell centres (median 43).

Sentinel-2 surface reflectance scenes of 2018 were linked to Cloud Score+ by scene identifier and masked where the cumulative distribution score (cs_cdf) was below 0.60. For every polygon and quarter, the area fraction with at least one clear observation was computed with the same reducer; it equalled 1.000 for all 10,144 polygons in all four quarters, so no polygon had an empty composite. Sentinel-1 ground range detected scenes (interferometric wide swath, VV and VH) were used as distributed in Earth Engine, that is, after thermal noise removal, radiometric calibration to backscatter coefficient and orthorectification with a digital elevation model, without radiometric terrain flattening, and expressed in decibels. Quarterly medians were computed per pixel from both orbit directions. Table S9 lists the scene counts; per grid cell, every polygon had at least 6 ascending and 11 descending observations in every quarter. Medians from a single orbit direction were evaluated as a sensitivity analysis (Table S5).

Table S1. Sequential selection of LUCAS 2018 Copernicus polygons

Selection step	Total	Spain	Greece
Catalogue (EU)	63,287	9,389	2,119
Spain and Greece	11,508	9,389	2,119
Quality screened subset	10,890	9,028	1,862
Classes B to F	10,859	9,005	1,854
Area of at least 400 m2	10,144	8,359	1,785

S2. Algorithms

Test block cluster bootstrap with resampled training draws (used for all intervals in the main text):

```
for each test set (Spain: G = 62 blocks of 50 km; Greece: G = 112 blocks of 50 km):
  C[r, g] = confusion matrix of repetition r restricted to test block g
  draw block weights w*[b, g] ~ Multinomial(G, 1/G), b = 1..2000   (shared by all models and budgets)
  for each budget n: draw repetition weights v*[b, r] ~ Multinomial(R, 1/R)
      (shared by all representations and classifiers of budget n; independent between budgets)
  for each representation and classifier at budget n:
      F*[b, r] = macro F1 of sum_g w*[b, g] C[r, g]
      F*[b]    = sum_r v*[b, r] F*[b, r] / R
  difference A - B: D*[b] = F*_A[b] - F*_B[b];  interval = 2.5th and 97.5th percentiles of D*
```

Intervals conditional on the training draws (block weights only, v* = 1) are archived with the results (columns ci_lo and ci_hi) and are on average about 25% narrower.

Dissimilarity index (Equation 9 of the main text):

```
standardise every feature with the mean and SD of the Spanish training pool (6,129 polygons)
d_train = median over pool samples of the Euclidean distance to the nearest OTHER pool sample (self excluded)
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```
for each test sample j: DI_j = Euclidean distance to the nearest pool sample / d_train
no feature weighting is applied; one nearest neighbour is used
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The index follows the idea of the area of applicability of Meyer and Pebesma (2021) but omits their weighting of predictors by variable importance and their threshold derived from cross-validation folds.

S3. Additional results

Table S2. Macro F1 score of Random Forest for all feature sets (mean ± standard deviation over 20 training repetitions; five seeds for the full pool)

Test set	Representation	n = 10	n = 25	n = 50	n = 100	Full pool
Spain	S2 annual	0.454 ± 0.032	0.491 ± 0.021	0.514 ± 0.019	0.538 ± 0.010	0.613 ± 0.003
Spain	S2 seasonal	0.473 ± 0.028	0.524 ± 0.013	0.549 ± 0.017	0.572 ± 0.009	0.651 ± 0.004
Spain	S1 + S2 seasonal	0.487 ± 0.023	0.534 ± 0.017	0.559 ± 0.012	0.583 ± 0.008	0.666 ± 0.003
Spain	S1 ascending + S2 seasonal	0.482 ± 0.021	0.528 ± 0.016	0.552 ± 0.015	0.575 ± 0.010	0.660 ± 0.003
Spain	S1 descending + S2 seasonal	0.482 ± 0.024	0.526 ± 0.014	0.552 ± 0.014	0.576 ± 0.009	0.659 ± 0.004
Spain	AE	0.541 ± 0.023	0.596 ± 0.015	0.617 ± 0.010	0.638 ± 0.008	0.695 ± 0.003
Spain	AE (unit norm)	0.542 ± 0.023	0.596 ± 0.014	0.617 ± 0.012	0.637 ± 0.009	0.694 ± 0.003
Spain	AE + S2 seasonal	0.531 ± 0.023	0.582 ± 0.016	0.604 ± 0.012	0.630 ± 0.010	0.707 ± 0.002
Greece	S2 annual	0.292 ± 0.035	0.303 ± 0.027	0.323 ± 0.022	0.336 ± 0.013	0.361 ± 0.002
Greece	S2 seasonal	0.309 ± 0.024	0.334 ± 0.025	0.360 ± 0.021	0.381 ± 0.018	0.445 ± 0.001
Greece	S1 + S2 seasonal	0.319 ± 0.031	0.347 ± 0.026	0.377 ± 0.023	0.393 ± 0.019	0.448 ± 0.005
Greece	S1 ascending + S2 seasonal	0.317 ± 0.029	0.343 ± 0.022	0.373 ± 0.021	0.390 ± 0.020	0.449 ± 0.005
Greece	S1 descending + S2 seasonal	0.313 ± 0.030	0.343 ± 0.026	0.372 ± 0.022	0.386 ± 0.018	0.436 ± 0.002
Greece	AE	0.400 ± 0.023	0.419 ± 0.026	0.438 ± 0.019	0.463 ± 0.013	0.466 ± 0.003
Greece	AE (unit norm)	0.402 ± 0.023	0.420 ± 0.025	0.439 ± 0.016	0.462 ± 0.017	0.465 ± 0.003
Greece	AE + S2 seasonal	0.378 ± 0.030	0.404 ± 0.023	0.424 ± 0.025	0.449 ± 0.018	0.483 ± 0.007

Table S3. Macro F1 score of multinomial logistic regression for all feature sets (mean ± standard deviation over 20 training repetitions; one fit for the full pool)

Test set	Representation	n = 10	n = 25	n = 50	n = 100	Full pool
Spain	S2 annual	0.479 ± 0.036	0.522 ± 0.017	0.542 ± 0.015	0.554 ± 0.009	0.595 ± 0.000
Spain	S2 seasonal	0.479 ± 0.032	0.531 ± 0.021	0.564 ± 0.015	0.582 ± 0.009	0.648 ± 0.000
Spain	S1 + S2 seasonal	0.493 ± 0.030	0.544 ± 0.021	0.576 ± 0.013	0.597 ± 0.008	0.675 ± 0.000
Spain	S1 ascending + S2 seasonal	0.487 ± 0.030	0.534 ± 0.023	0.571 ± 0.017	0.590 ± 0.009	0.679 ± 0.000
Spain	S1 descending + S2 seasonal	0.490 ± 0.028	0.540 ± 0.021	0.569 ± 0.015	0.594 ± 0.008	0.666 ± 0.000
Spain	AE	0.548 ± 0.035	0.589 ± 0.021	0.614 ± 0.018	0.638 ± 0.010	0.722 ± 0.000
Spain	AE (unit norm)	0.547 ± 0.035	0.589 ± 0.020	0.615 ± 0.017	0.638 ± 0.010	0.724 ± 0.000
Spain	AE + S2 seasonal	0.549 ± 0.033	0.589 ± 0.022	0.615 ± 0.019	0.640 ± 0.009	0.733 ± 0.000
Greece	S2 annual	0.292 ± 0.022	0.315 ± 0.015	0.337 ± 0.015	0.340 ± 0.012	0.348 ± 0.000
Greece	S2 seasonal	0.325 ± 0.033	0.364 ± 0.034	0.398 ± 0.022	0.425 ± 0.023	0.454 ± 0.000
Greece	S1 + S2 seasonal	0.330 ± 0.036	0.369 ± 0.031	0.410 ± 0.024	0.421 ± 0.026	0.448 ± 0.000
Greece	S1 ascending + S2 seasonal	0.331 ± 0.032	0.366 ± 0.036	0.410 ± 0.026	0.425 ± 0.027	0.460 ± 0.000
Greece	S1 descending + S2 seasonal	0.320 ± 0.033	0.365 ± 0.034	0.402 ± 0.024	0.416 ± 0.029	0.437 ± 0.000
Greece	AE	0.414 ± 0.035	0.425 ± 0.030	0.447 ± 0.038	0.483 ± 0.025	0.538 ± 0.000
Greece	AE (unit norm)	0.415 ± 0.035	0.425 ± 0.031	0.447 ± 0.037	0.482 ± 0.024	0.540 ± 0.000

Greece	AE + S2 seasonal	0.408 ± 0.036	0.425 ± 0.028	0.457 ± 0.040	0.487 ± 0.023	0.592 ± 0.000

Note: Logistic regression is deterministic, so the full pool row has no variation.

Table S4. Paired macro F1 differences with 95% two-level bootstrap intervals

Classifier	Comparison	Test set	n = 10	n = 25	n = 50	n = 100	Full pool
RF	AE − S2 seasonal	Spain	0.068 (0.051, 0.088)	0.072 (0.057, 0.089)	0.069 (0.051, 0.087)	0.066 (0.049, 0.084)	0.043 (0.018, 0.071)
RF	AE − S2 seasonal	Greece	0.091 (0.060, 0.120)	0.086 (0.052, 0.117)	0.079 (0.047, 0.108)	0.083 (0.050, 0.113)	0.021 (−0.006, 0.047)
RF	AE − S1 + S2 seasonal	Spain	0.054 (0.038, 0.072)	0.063 (0.048, 0.079)	0.058 (0.042, 0.075)	0.055 (0.040, 0.071)	0.029 (0.003, 0.057)
RF	AE − S1 + S2 seasonal	Greece	0.081 (0.052, 0.108)	0.073 (0.043, 0.103)	0.061 (0.032, 0.088)	0.070 (0.040, 0.098)	0.018 (−0.008, 0.042)
LR	AE − S2 seasonal	Spain	0.068 (0.050, 0.086)	0.058 (0.039, 0.079)	0.051 (0.032, 0.071)	0.056 (0.036, 0.079)	0.073 (0.049, 0.103)
LR	AE − S2 seasonal	Greece	0.089 (0.058, 0.119)	0.062 (0.033, 0.091)	0.049 (0.016, 0.083)	0.058 (0.027, 0.088)	0.084 (0.045, 0.130)
LR	AE − S1 + S2 seasonal	Spain	0.054 (0.038, 0.070)	0.045 (0.027, 0.063)	0.039 (0.020, 0.058)	0.041 (0.024, 0.061)	0.046 (0.023, 0.071)
LR	AE − S1 + S2 seasonal	Greece	0.084 (0.056, 0.110)	0.056 (0.031, 0.082)	0.037 (0.008, 0.066)	0.062 (0.036, 0.089)	0.090 (0.051, 0.134)

Table S5. Sensitivity analyses with Random Forest: contribution of Sentinel-1, single orbit direction and unit-length embeddings (paired differences with 95% two-level bootstrap intervals)

Comparison (RF)	Test set	n = 10	n = 25	n = 50	n = 100	Full pool
S1 + S2 seasonal − S2 seasonal	Spain	0.014 (0.005, 0.023)	0.009 (0.002, 0.016)	0.011 (0.003, 0.019)	0.011 (0.003, 0.017)	0.015 (0.002, 0.028)
S1 + S2 seasonal − S2 seasonal	Greece	0.010 (0.000, 0.020)	0.013 (0.004, 0.023)	0.018 (0.008, 0.028)	0.013 (0.001, 0.024)	0.003 (−0.009, 0.015)
S1 ascending + S2 seasonal − S1 + S2 seasonal	Spain	−0.005 (−0.011, 0.001)	−0.005 (−0.010, 0.000)	−0.008 (−0.013, −0.002)	−0.008 (−0.014, −0.002)	−0.006 (−0.016, 0.003)
S1 ascending + S2 seasonal − S1 + S2 seasonal	Greece	−0.002 (−0.009, 0.006)	−0.003 (−0.011, 0.004)	−0.005 (−0.012, 0.002)	−0.004 (−0.012, 0.005)	0.001 (−0.009, 0.011)
S1 descending + S2 seasonal − S1 + S2 seasonal	Spain	−0.005 (−0.012, 0.001)	−0.007 (−0.013, −0.001)	−0.007 (−0.013, −0.001)	−0.007 (−0.013, −0.001)	−0.007 (−0.016, 0.002)
S1 descending + S2 seasonal − S1 + S2 seasonal	Greece	−0.006 (−0.013, 0.001)	−0.004 (−0.010, 0.002)	−0.006 (−0.013, 0.001)	−0.007 (−0.015, 0.000)	−0.013 (−0.023, −0.003)
AE (unit norm) − AE	Spain	0.001 (−0.002, 0.004)	0.000 (−0.004, 0.003)	−0.001 (−0.004, 0.003)	−0.002 (−0.005, 0.002)	−0.001 (−0.008, 0.007)
AE (unit norm) − AE	Greece	0.001 (−0.003, 0.006)	0.000 (−0.005, 0.005)	0.001 (−0.005, 0.007)	−0.002 (−0.009, 0.005)	−0.001 (−0.007, 0.005)

Table S6. Comparisons across label budgets (paired differences with 95% two-level bootstrap intervals; repetitions of different budgets resampled independently)

Test set	Classifier	Comparison	Difference (95% interval)
Spain	RF	AE, n = 25 vs S2 seasonal, n =	0.024 (0.007, 0.041)

		100	
Spain	LR	AE, n = 25 vs S2 seasonal, n = 100	0.007 (−0.012, 0.028)
Spain	RF	AE, n = 25 vs S1 + S2 seasonal, n = 100	0.013 (−0.003, 0.029)
Spain	LR	AE, n = 25 vs S1 + S2 seasonal, n = 100	−0.007 (−0.026, 0.013)
Spain	RF	AE, n = 10 vs S2 seasonal, n = 50	−0.008 (−0.027, 0.010)
Spain	LR	AE, n = 10 vs S2 seasonal, n = 50	−0.016 (−0.040, 0.006)
Greece	RF	AE, n = 10 vs S2 seasonal, n = 100	0.020 (−0.009, 0.047)
Greece	LR	AE, n = 10 vs S2 seasonal, n = 100	−0.011 (−0.042, 0.020)
Greece	RF	AE, n = 25 vs S1 + S2 seasonal, n = 100	0.026 (−0.002, 0.052)
Greece	LR	AE, n = 25 vs S1 + S2 seasonal, n = 100	0.005 (−0.023, 0.032)

Table S7. Test and training set sizes and distances in the validation sensitivity analysis (Random Forest, median over ten repetitions)

Protocol	Median test size	Median training pool	Median distance to nearest training sample (km)
Random	2,090	6,269	4.0
10 km blocks	2,085	6,274	6.0
25 km blocks	2,116	6,244	8.9
50 km blocks	2,108	6,252	13.7
100 km blocks	2,103	6,256	22.0

Note: The sensitivity analysis uses no buffer and redraws the held out samples or blocks in every repetition; it is therefore not identical to the locked holdout.

Table S8. Dissimilarity of test samples to the Spanish training pool

Representation	Test set	Median DI	Share with DI > 2
S2 annual	Spain	0.99	4.2%
S2 annual	Greece	0.92	1.5%
S2 seasonal	Spain	1.01	3.0%
S2 seasonal	Greece	1.02	3.7%
S1 + S2 seasonal	Spain	1.01	1.7%
S1 + S2 seasonal	Greece	1.00	2.3%
AE	Spain	1.01	0.0%
AE	Greece	1.38	0.4%

Table S9. Sentinel-1 interferometric wide swath scenes with VV and VH polarisation intersecting the study region in 2018, by quarter and orbit direction

Quarter	Ascending scenes	Descending scenes
1	2014	2026
2	1876	1992
3	2724	2889
4	3121	3321

S4. Additional figures

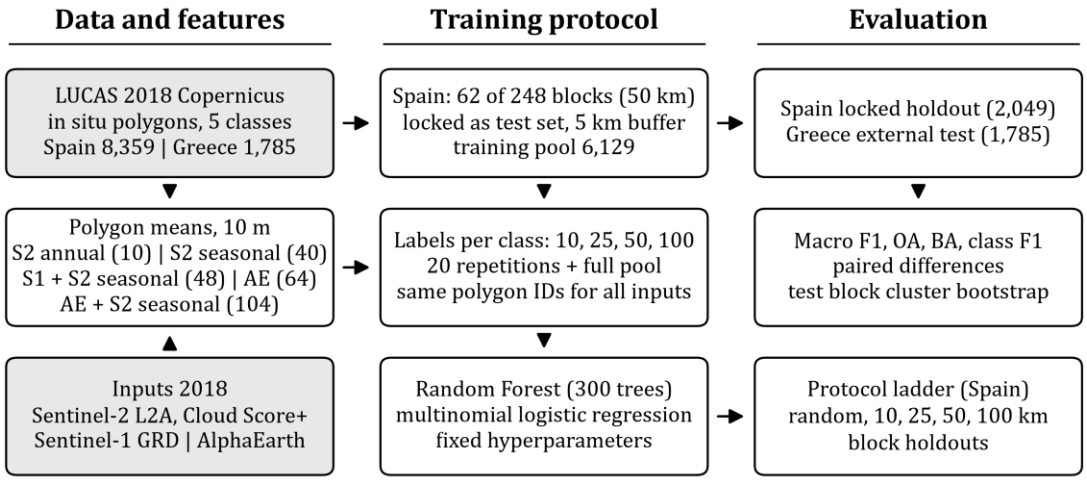


Figure S1. Workflow of data preparation, training and evaluation.

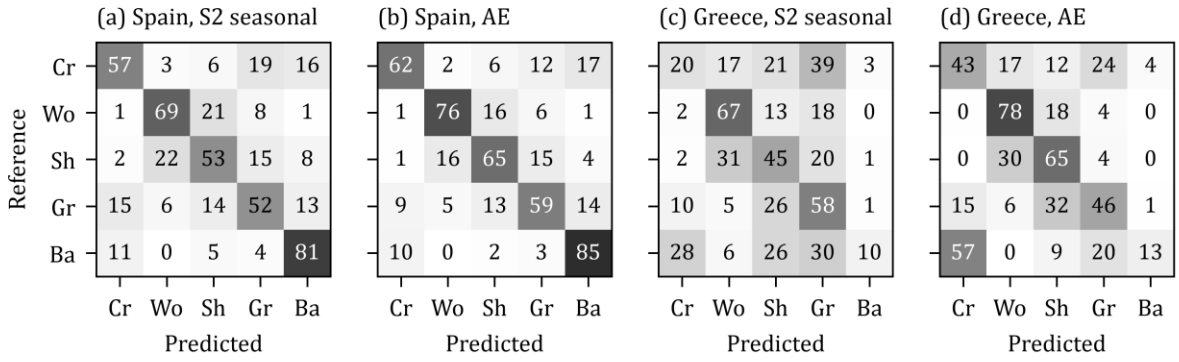


Figure S2. Row normalised confusion matrices (%) of Random Forest with 100 labels per class, pooled over the 20 training repetitions (the pooled counts are not independent samples). Supports: Spain 749, 485, 305, 376, 134; Greece 511, 468, 205, 545, 56 (cropland, woodland, shrubland, grassland, bare land). Cr, cropland; Wo, woodland; Sh, shrubland; Gr, grassland; Ba, bare land.

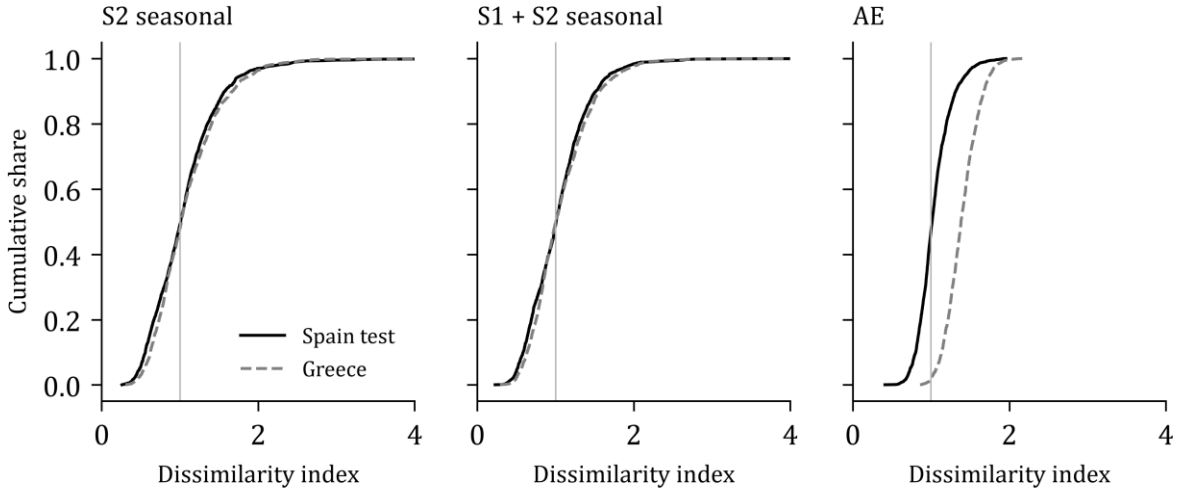


Figure S3. Cumulative distributions of the dissimilarity index of Spanish test and Greek polygons for three representations; the vertical line marks DI = 1.

S5. Data and code archive

The archive contains the configuration, all scripts and the following files: the reference and feature tables extracted from Earth Engine (data/raw), the analysis table (data/processed/samples.parquet), the split identifier lists with SHA-256 checksums and the 50 km test block geometries (data/splits), per run metrics (outputs/metrics/e1_runs.csv, e2_ladder.csv), a long format prediction table with one row per test sample and run (outputs/metrics/predictions_long.parquet; columns run_id, test_set, sample_id, point_id, block_id, true_label, prediction, representation, classifier, budget, repetition, rf_seed, sampling_seed), and the bootstrap outputs. The command python reproduce_all.py regenerates all tables and figures from the archived feature tables; python reproduce_all.py --extract repeats the Earth Engine extraction. Software: Python 3.12.10, scikit-learn 1.9.1, earthengine-api 1.7.46 (full lock file in reports/requirements-lock.txt).

Split checksums: train_pool_spain_ids.csv: 6129 polygons, SHA-256 81238f623f01b548…; test_spain_ids.csv: 2049 polygons, SHA-256 4aac5ba2690b4d58…; buffer_excluded_spain_ids.csv: 181 polygons, SHA-256 9ee1b7c09d9fff11…; test_greece_ids.csv: 1785 polygons, SHA-256 4ad890a0a8282b42….